Threat Intelligence Agent (Agent 5)

Deep Technical Specification & Operational Manual

Executive Overview & Purpose

The Threat Intelligence Agent queries vulnerability databases, maps attack techniques to the MITRE ATT&CK; Enterprise matrix, assesses weaponization risk, and generates production-ready detection signatures (Sigma, YARA, Splunk SPL).

1. Architectural Overview & Workflow

• Position in LangGraph Engine: Primary node in Pipeline C (Threat Intel) and third node in Pipeline B.

• Input Processing: Extracts CVE identifiers (CVE-YYYY-NNNN) and attack descriptions from state.

• State Propagation: Writes state.threat_intel_report containing CVE CVSS scores, MITRE mappings, and generated detection rules.

2. National Vulnerability Database (NVD 2.0) Integration

• NVD REST API 2.0: Queries https://services.nvd.nist.gov/rest/json/cves/2.0?cveId={cve_id} asynchronously.

• Extracted Metrics: Retrieves CVSS v3.1 base score, severity rating, vector string (e.g. CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:H/I:H/A:H), and published timestamps.

3. MITRE ATT&CK; Enterprise Matrix Mapping

• Local STIX Dataset: Queries Enterprise ATT&CK; matrix for tactics and technique IDs (e.g., T1190 Exploit Public-Facing Application, T1059 Command & Scripting Interpreter).

• Context Alignment: Aligns mapped techniques with incident triage findings.

4. Automated Detection Rule Generation Engine

• Sigma Rules: Generates valid YAML Sigma detection rules for SIEM deployment.

• YARA Rules: Generates memory and disk scanning YARA rules with hex/string patterns.

• Splunk SPL: Generates ready-to-use search queries for SOC monitoring dashboards.